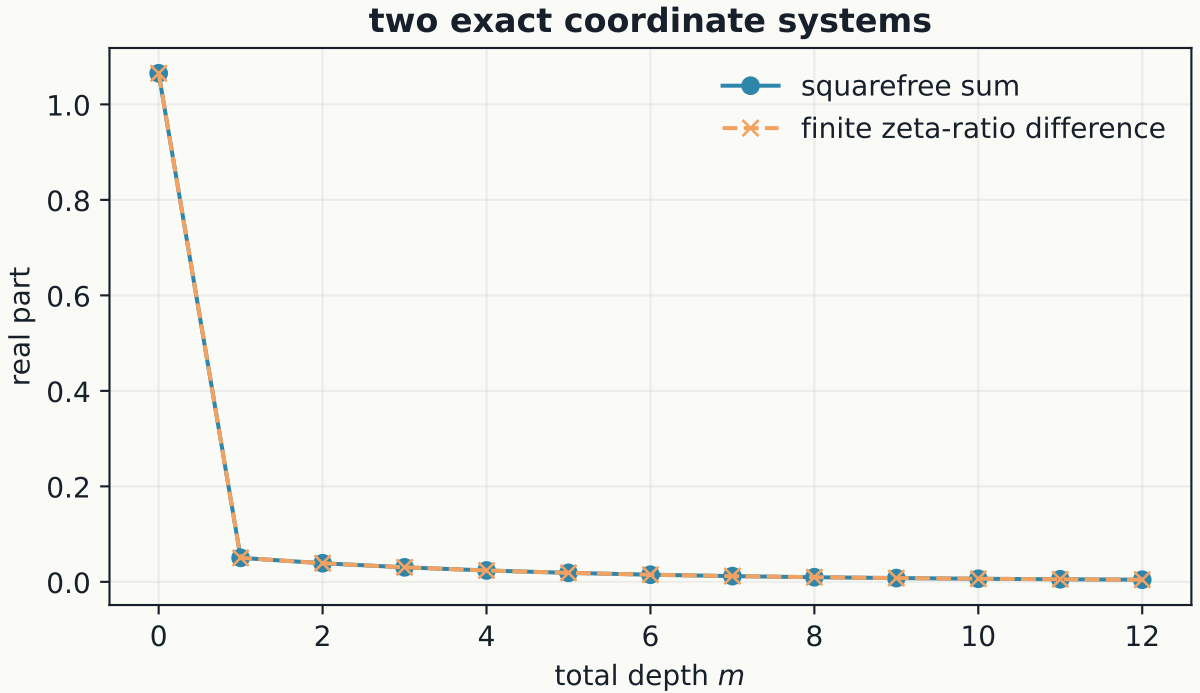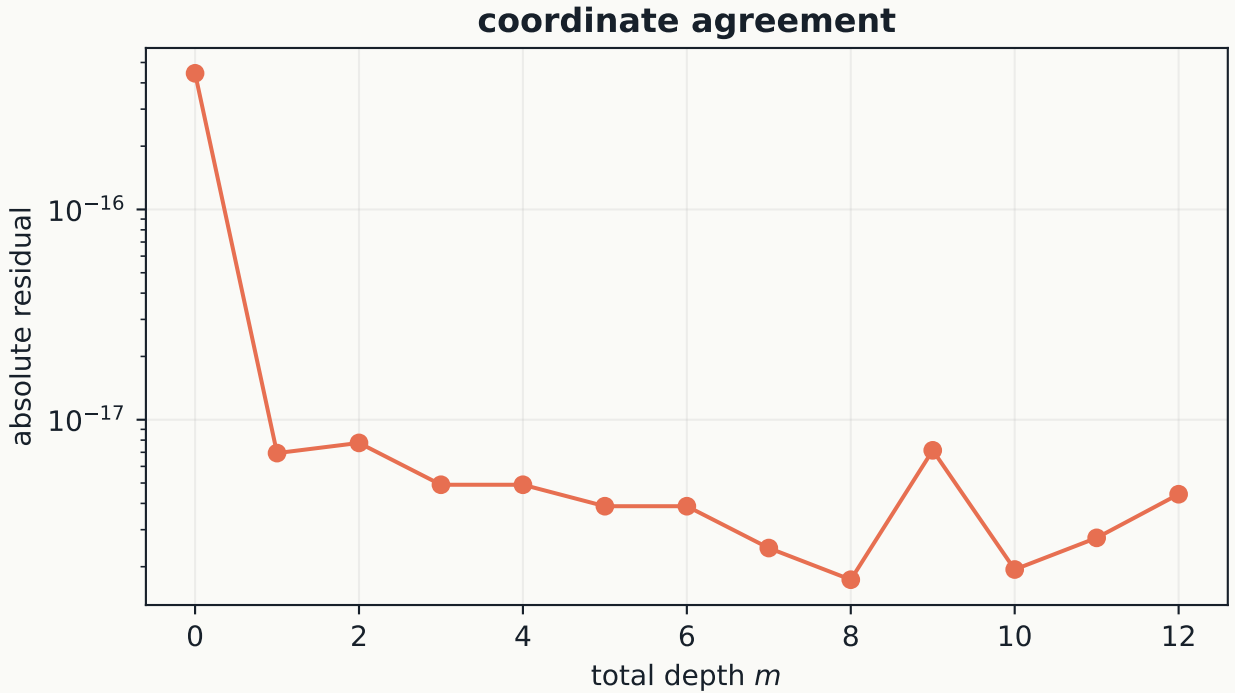The kernel is a finite difference of a squarefree L-ratio